

Exercise 2.4

1. Without using a calculator, evaluate the following:

(i) $\log_2 18 - \log_2 9$

$$\begin{aligned} & \log_2 18 - \log_2 9 \\ &= \log_2 \frac{18}{9} \\ &= \log_2 2 \\ &= 1 \quad \because \log_a a = 1 \end{aligned}$$

(ii) $\log_2 64 - \log_2 2$

$$\begin{aligned} & \log_2 64 - \log_2 2 \\ &= \log_2 64 \times 2 \\ &= \log_2 128 \\ &= \log_2 2^7 \\ &= 7 \log_2 2 \\ &= 7(1) \quad \because \log_a a = 1 \\ &= 7 \end{aligned}$$

(iii) $\frac{1}{3} \log_3 8 - \log_3 18$

$$\begin{aligned} & \frac{1}{3} \log_3 8 - \log_3 18 \\ &= \frac{1}{3} \log_3 2^3 - \log_3 2 \cdot 3^2 \\ &= \frac{1}{3} \log_3 2 - [\log_3 2 + \log_3 3^2] \\ &= \frac{1}{3} \log_3 2 - \log_3 2 - 2 \log_3 3 \\ &= -2 \log_3 3 \\ &= -2(1) \quad \because \log_a a = 1 \\ &= -2 \end{aligned}$$

(iv) $2 \log 2 + \log 25$

$$\begin{aligned} & 2 \log 2 + \log 25 \\ &= 2 \log 2 + \log 5^2 \\ &= 2 \log 2 + 2 \log 5 \\ &= 2[\log 2 + \log 5] \\ &= 2 \log 2 \times 5 \\ &= 2 \log 10 \\ &= 2 \log_{10} 10 \\ &= 2(1) \\ &= 2 \end{aligned}$$

(v) $\frac{1}{3} \log_3 64 + 2 \log_5 25$

$$\begin{aligned} & \frac{1}{3} \log_4 64 + 2 \log_5 25 \\ &= \frac{1}{3} \log_4 4^3 + 2 \log_5 5^2 \\ &= \frac{3}{3} \log_4 4 + 2 \times 2 \log_5 5 \\ &= \log_4 4 + 4 \log_5 5 \\ &= 1 + 4(1) \end{aligned}$$

$$\begin{aligned} &= 1 + 4 \\ &= 5 \end{aligned}$$

(vi) $\log_3 12 + \log_3 0.25$

$$\begin{aligned} & \log_3 12 + \log_3 0.25 \\ &= \log_3 (12 \times 0.25) \\ &= \log_3 \left(12 \times \frac{25}{100} \right) \\ &= \log_3 3 \\ &= 1 \end{aligned}$$

2. Write the following as a single logarithm:

(i) $\frac{1}{2} \log 25 - 2 \log 3$

$$\begin{aligned} & \frac{1}{2} \log 25 - 2 \log 3 \\ &= \log (5^2)^{\frac{1}{2}} - \log 3^2 \\ &= \log 5 - \log 9 \\ &= \log \frac{5}{9} \end{aligned}$$

(ii) $\log 9 - \log \frac{1}{3}$

$$\begin{aligned} & \log 9 - \log \frac{1}{3} \\ &= \log \frac{9}{1/3} \\ &= \log \frac{9 \times 3}{1} \\ &= \log 27 \end{aligned}$$

(iii) $\log_5 b^2 \cdot \log_a 5^3$

$$\begin{aligned} & \log_5 b^2 \cdot \log_a 5^3 \\ &= 2 \log_5 b \cdot 3 \log_a 5 \\ &= 6 \log_5 b \cdot \log_a 5 \\ &= 6 \frac{\log b}{\log 5} \cdot \frac{\log 5}{\log a} \quad \because \log_b x = \frac{\log_a x}{\log_a b} \\ &= 6 \frac{\log b}{\log a} \\ &= 6 \log_a b \quad \because \frac{\log_a x}{\log_a b} = \log_b x \end{aligned}$$

(iv) $2 \log_3 x + \log_3 y$

$$\begin{aligned} & 2 \log_3 x + \log_3 y \\ &= \log_3 x^2 + \log_3 y \\ &= \log_3 x^2 y \end{aligned}$$

(v) $4 \log_5 x - \log_5 y + \log_5 z$

$$\begin{aligned} & 4 \log_5 x - \log_5 y + \log_5 z \\ &= \log_5 x^4 - \log_5 y + \log_5 z \\ &= \log_5 \frac{x^4 z}{y} \end{aligned}$$

(vi) $2 \ln a + 3 \ln b - 4 \ln c$

$$2 \ln a + 3 \ln b - 4 \ln c$$

$$= \ln a^2 + \ln b^3 - \ln c^4$$

$$= \ln \frac{a^2 b^3}{c^4}$$

3. Expand the following using laws of logarithms:

(i) $\log\left(\frac{11}{5}\right)$

$$\log\left(\frac{11}{5}\right)$$

$$= \log 11 - \log 5$$

(ii) $\log_5 \sqrt{8a^6}$

$$\ln \frac{a^2 b}{c}$$

$$= \ln a^2 + \ln b - \ln c$$

$$= 2 \ln a + \ln b - \ln c$$

(iii) $\ln \frac{a^2 b}{c}$

$$\ln \frac{a^2 b}{c}$$

$$= \ln a^2 + \ln b - \ln c$$

$$= 2 \ln a + \ln b - \ln c$$

(iv) $\log\left(\frac{xy}{z}\right)^{\frac{1}{9}}$

$$\log\left(\frac{xy}{z}\right)^{\frac{1}{9}}$$

$$= \frac{1}{9} \left[\log\left(\frac{xy}{z}\right) \right]$$

$$= \frac{1}{9} [\log x + \log y - \log z]$$

(v) $\ln \sqrt[3]{16x^3}$

$$\ln \sqrt[3]{16x^3}$$

$$= \ln(16x^3)^{\frac{1}{3}}$$

$$= \frac{1}{3} [\ln 2^4 x^3]$$

$$= \frac{1}{3} [\ln 2^4 + \ln x^3]$$

$$= \frac{1}{3} [4 \ln 2 + 3 \ln x]$$

$$= \frac{4}{3} \ln 2 + \ln x$$

$$= \frac{4}{3} \ln 2 + \ln x$$

(vi) $\log_2 \left(\frac{1-a}{b}\right)^5$

$$\log_2 \left(\frac{1-a}{b}\right)^5$$

$$= 5 \log_2 \left(\frac{1-a}{b}\right)$$

$$= 5 [\log_2 (1-a) - \log_2 b]$$

4. Find the value of x in the following equations:

(i) $\log 2 + \log x = 1$

$$\log 2 + \log x = 1$$

$$\log 2x = 1$$

$$\log_{10} 2x = 1$$

$$10^1 = 2x$$

$$\frac{10}{2} = x$$

$$5 = x$$

$$x = 5$$

(ii) $\log_2 x + \log_2 8 = 5$

$$\log_2 x + \log_2 8 = 5$$

$$\log_2 8x = 5$$

$$2^5 = 8x$$

$$32 = 8x$$

$$\frac{32}{8} = x$$

$$4 = x$$

$$x = 4$$

(iii) $(81)^x = (243)^{x+2}$

$$(81)^x = (243)^{x+2}$$

$$(3^4)^x = (3^5)^{x+2}$$

$$3^{4x} = 3^{5x+10}$$

$$\Rightarrow 4x = 5x + 10$$

$$4x - 5x = 10$$

$$-x = 10$$

$$x = -10$$

(iv) $\left(\frac{1}{27}\right)^{x-6} = 27$

$$\left(\frac{1}{27}\right)^{x-6} = 27$$

$$(27^{-1})^{x-6} = 27$$

$$27^{-x+6} = 27^1$$

$$\Rightarrow -x + 6 = 1$$

$$-x = 1 - 6$$

$$-x = -5$$

$$x = 5$$

(v) $\log(5x - 10) = 2$

$$\log(5x - 10) = 2$$

$$\log_{10}(5x - 10) = 2$$

$$10^2 = 5x - 10$$

$$100 = 5x - 10$$

$$100 + 10 = 5x$$

$$110 = 5x$$

$$\frac{110}{5} = x$$

$$22 = x$$

$$x = 22$$

$$(vi) \log_2(x+1) - \log_2(x-4) = 2$$

$$\log_2(x+1) - \log_2(x-4) = 2$$

$$\log_2\left(\frac{x+1}{x-4}\right) = 2$$

$$2^2 = \frac{x+1}{x-4}$$

$$4 = \frac{x+1}{x-4}$$

$$4(x-4) = x+1$$

$$4x - 16 = x + 1$$

$$4x - x = 1 + 16$$

$$3x = 17$$

$$x = \frac{17}{3}$$

$$x = 5\frac{2}{3}$$

5. Find the values of the following with the help of logarithm table:

$$(i) \frac{3.68 \times 4.21}{5.234}$$

Let

$$x = \frac{3.68 \times 4.21}{5.234}$$

$$\log x = \log\left(\frac{3.68 \times 4.21}{5.234}\right)$$

$$\log x = \log 3.68 + \log 4.21 - \log 5.234$$

$$\log x = 0.5658 + 0.6243 - 0.7188$$

$$\log x = 0.4713$$

$$\text{Antilog}(\log x) = \text{Anti log}(0.4713)$$

$$x = 2.960$$

$$(ii) 4.67 \times 2.11 \times 2.397$$

Let

$$x = 4.67 \times 2.11 \times 2.397$$

$$\log x = \log(4.67 \times 2.11 \times 2.397)$$

$$\log x = \log 4.67 + \log 2.11 + \log 2.397$$

$$\log x = 0.6693 + 0.3243 + 0.3797$$

$$\log x = 1.3733$$

$$\text{Antilog}(\log x) = \text{Anti log}(1.3733)$$

$$x = 23.62$$

$$(iii) \frac{(20.46)^2 \times (2.4122)}{754.3}$$

Let

$$x = \frac{(20.46)^2 \times (2.4122)}{754.3}$$

$$\log x = \log\left[\frac{(20.46)^2 \times (2.4122)}{754.3}\right]$$

$$\log x = \log(20.46)^2 + \log 2.4122 - \log 754.3$$

$$\log x = 2 \log 20.46 + \log 2.4122 - \log 754.3$$

$$\log x = 2(1.3109) + 0.3824 - 2.8775$$

$$\log x = 2.6218 + 0.3824 - 2.8775$$

$$\log x = 0.1267$$

$$\text{Antilog}(\log x) = \text{Anti log}(0.1267)$$

$$x = 1.338$$

$$(iv) \frac{\sqrt[3]{9.364} \times 21.64}{3.21}$$

Let

$$x = \frac{\sqrt[3]{9.364} \times 21.64}{3.21}$$

$$\begin{aligned}
\log x &= \log \left[\frac{\sqrt[3]{9.364} \times 21.64}{3.21} \right] \\
\log x &= \log \sqrt[3]{9.364} + \log 21.64 - \log 3.21 \\
\log x &= \log (9.364)^{1/3} + \log 21.64 - \log 3.21 \\
\log x &= \frac{1}{3} \log 9.364 + \log 21.64 - \log 3.21 \\
\log x &= \frac{1}{3} (0.9715) + 1.335 - 0.5065 \\
\log x &= 0.3238 + 1.335 - 0.5065 \\
\log x &= 1.1523 \\
\text{Antilog}(\log x) &= \text{Anti log}(1.1523) \\
x &= \mathbf{14.20}
\end{aligned}$$

6. The formula to measure the magnitude of earthquakes is given by $M = \log_{10} \left(\frac{A}{A_0} \right)$. If amplitude (A) is 10,000 and reference amplitude (A_0) is 10, what is the magnitude of the earthquake?

Since

$$\begin{aligned}
M &= \log_{10} \left(\frac{A}{A_0} \right) \\
\text{Put } A &= 10000 \text{ and } A_0 = 10 \\
M &= \log_{10} \left(\frac{10000}{10} \right) \\
M &= \log_{10} (1000) \\
M &= \log_{10} 10^3 \\
M &= 3 \log_{10} 10 \\
M &= 3(1) \\
M &= \mathbf{3}
\end{aligned}$$

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7. Abdullah invested Rs. 100,000 in a saving scheme and gains interest at the rate of 5% per annum so that the total value of this investment after t years is Rs y . This is modelled by an equation $y = 1,00,000(1.05)^t$, $t \geq 0$. Find after how many years the investment will be doubled?

$$\begin{aligned}
\text{Initial investment amount} &= 100000 \text{ Rs.} \\
\text{Double the investment} &= y = 200000 \text{ Rs} \\
\text{Time} &= t = ?
\end{aligned}$$

Since

$$\begin{aligned}
y &= 100000(1.05)^t \\
200000 &= 100000(1.05)^t \\
\frac{200000}{100000} &= (1.05)^t \\
2 &= (1.05)^t \\
\log 2 &= \log (1.05)^t \\
\log 2 &= t \log 1.05 \\
\frac{\log 2}{\log 1.05} &= t \\
\frac{0.3010}{0.0212} &= t \\
14.2 &= t \\
t &= \mathbf{14.2 \text{ years}}
\end{aligned}$$

8. Huria is hiking up a mountain where the temperature (T) decreases by 3% (or a factor of 0.97) for every 100 meters gained in altitude. The initial temperature (T_i) at sea level is 20°C . Using the formula $T = T_i \times 0.97^{h/100}$, calculate the temperature at an altitude (h) of 500 meters.

$$\text{Initial Temperature} = T_i = 20^{\circ}\text{C}$$

$$\text{Altitude} = h = 500 \text{ m}$$

$$\text{Final Temperature} = T = ?$$

Since

$$T = T_i \times 0.97^{h/100}$$

$$\text{Put } T_i = 20 \text{ and } h = 500$$

$$T = 20 \times 0.97^{500/100}$$

$$T = 20 \times 0.97^5$$

$$\log T = \log[20 \times 0.97^5]$$

$$\log T = \log 20 + \log 0.97^5$$

$$\log T = \log 20 + 5 \log 0.97$$

$$\log T = 1.3010 + 5(-0.0132)$$

$$\log T = 1.3010 - 0.066$$

$$\log T = 1.235$$

$$\text{Anti log}(\log T) = \text{Anti log } 1.235$$

$$T = 17.18^{\circ}\text{C}$$

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